

## TrackGenesis FSD Internship – October 2025

### Project Title:

Smart Budget & Expense Planner

### Introduction

The Smart Budget & Expense Planner is a full-stack MERN (MongoDB, Express.js, React.js, Node.js) web application developed as part of the TrackGenesis Full Stack Development Internship. The platform helps users efficiently manage their personal finances by tracking income and expenses, setting monthly budgets, and receiving AI-powered financial suggestions. The application focuses on simplifying money management through data visualization, category-wise spending analytics, and real-time budget insights. Additionally, it supports user authentication, role-based access, and secure OTP-based verification. The ultimate goal of the project is to enable users to understand and control their financial habits using smart technology.

### Tech Stack

Frontend Technologies: React.js, Vite, HTML5, CSS3, , JavaScript, Chart.js

Backend Technologies: Node.js, Express.js

Database: MongoDB (NoSQL)

APIs and Services: Nodemailer, JWT, Gemini, Chart.js, AI Module

Development Tools: Visual Studio Code, GitHub, Postman

## Folder Structure

project-root/

|

├─ client/

| └─ public/

| └─ vite.svg

| └─ src/

| └─ assets/

| └─ budgetPlanning.jpg

| └─ chart.png

| └─ expense.png

| └─ goal.jpg

| └─ piggyicon.png

| └─ components/

| └─ AuthDialog.jsx

| └─ ProtectedRoute.jsx

| └─ Sidebar.jsx

| └─ layouts/

| └─ pages/

| └─ AdminDashboard.jsx

| └─ Authsuccess.jsx

| └─ BudgetPage.jsx

| └─ ExpenseDetails.jsx

| └─ Home.jsx

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| | | └─ IncomeDetails.jsx
| | | └─ Navbar.jsx
| | | └─ Settings.jsx
| | | └─ UserDashboard.jsx
| | └─ App.jsx
| | └─ main.jsx
| | └─ index.css
| └─ package.json
|
└─ server/
    └─ config/
        └─ db.js
        └─ Passport.js
    └─ controllers/
        └─ aicontroller.js
        └─ budgetController.js
        └─ expenseController.js
        └─ ...
    └─ middleware/
        └─ authMiddleware.js
    └─ models/
        └─ Budget.js
        └─ Expense.js
        └─ Income.js
        └─ PendingUser.js
```

```
|   |   └─ User.js
|   └─ routes/
|   |   └─ BudgetRoutes.js
|   |   └─ ExpenseRoutes.js
|   |   └─ IncomeRoutes.js
|   |   └─ authRoutes.js
|   |   └─ dashboardRoute.js
|   |   └─ forgetPassword.js
|   └─ utils/
|   |   └─ admin.js
|   |   └─ generateotp.js
|   |   └─ sendemail.js
|   └─ app.js
|   └─ package.json
|
└─ README.md
```

## Used Packages / Dependencies

Backend Dependencies: express, mongoose, bcryptjs, jsonwebtoken, nodemailer, cors, dotenv, passport, chart.js

Frontend Dependencies: react, react-router-dom, axios, tailwindcss, react-toastify, chart.js, vite

## Key Features & Functionalities

- User Authentication & Security: JWT authentication, OTP verification, password reset via email, and role-based access.
- Dashboard & Reports: Interactive charts for expenses and income, monthly summaries, and AI-generated insights.

- Expense Management: Add/edit/delete transactions, categorize expenses, manage payment modes, and set budgets.
- Reports & Downloads: Export CSV/PDF reports, apply filters, and generate monthly AI summaries.

### Challenges / Difficulties Faced

- AI-based Spending Suggestions: Building logic to analyze past transactions and provide personalized saving tips.
- Secure OTP Email Verification: Handling multiple OTP scenarios and ensuring validation and expiry management.
- Chart Visualization: Ensuring accurate data updates and state management using React hooks and context API.

### Conclusion

The Smart Budget & Expense Planner project provided a hands-on experience in developing a real-world full-stack web application using the MERN stack. This internship enhanced technical proficiency in backend API development, state management, authentication, and UI/UX implementation. It improved problem-solving, debugging, and teamwork skills, laying a strong foundation for a career in full-stack web development.